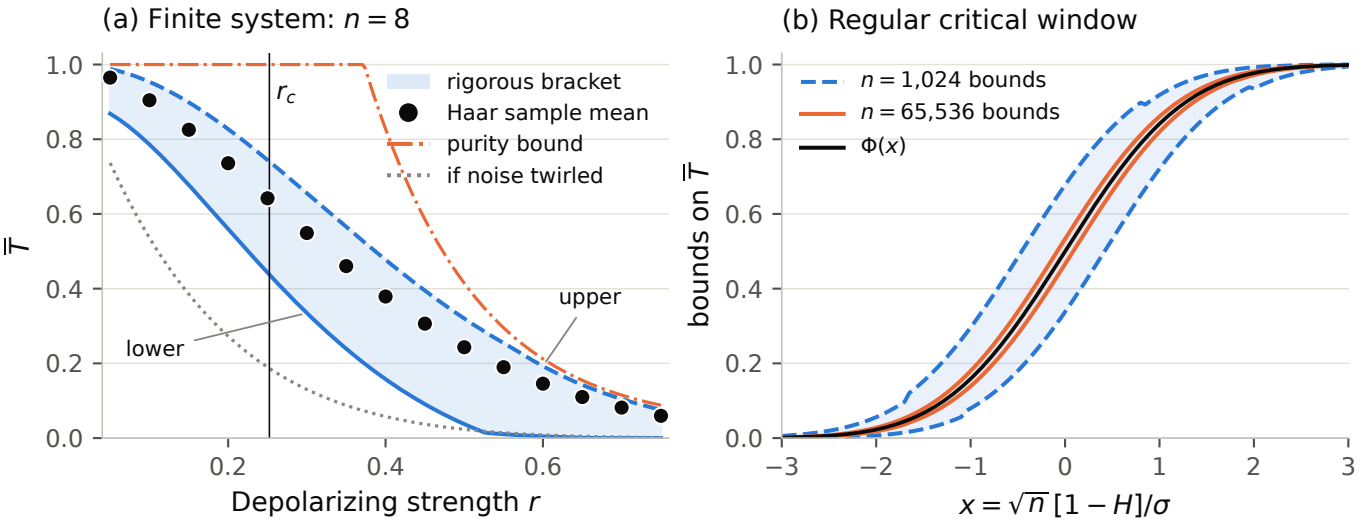
(a) Finite system: $n = 8$
rigorous bracket
Haar sample mean
purity bound
if noise twirled
r_c
upper
lower
$\overline{T}$
Depolarizing strength r
(b) Regular critical window
$n = 1{,}024$ bounds
$n = 65{,}536$ bounds
$\Phi(x)$
bounds on $\overline{T}$
$x = \sqrt{n}\,[1 - H]/\sigma$